will make the number of ones and zeros slightly different each time the signal is generated. The *probabilities* of the underlying process are constant, but the *statistics* of the acquired signal change each time the experiment is repeated. This random irregularity found in actual data is called by such names as: **statistical variation**, **statistical fluctuation**, and **statistical noise.**

This presents a bit of a dilemma. When you see the terms: *mean* and *standard deviation*, how do you know if the author is referring to the statistics of an actual signal, or the probabilities of the underlying process that created the signal? Unfortunately, the only way you can tell is by the context. This is not so for all terms used in statistics and probability. For example, the *histogram* and *probability mass function* (discussed in the next section) are matching concepts that are given separate names.

Now, back to Eq. 2-2, calculation of the standard deviation. As previously mentioned, this equation divides by N-1 in calculating the average of the squared deviations, rather than simply by N. To understand why this is so, imagine that you want to find the mean and standard deviation of some *process* that generates signals. Toward this end, you acquire a signal of N samples from the process, and calculate the mean of the signal via Eq. 2.1. You can then use this as an *estimate* of the mean of the underlying process; however, you know there will be an error due to statistical noise. In particular, for random signals, the typical error between the mean of the N points, and the mean of the underlying process, is given by:

EQUATION 2-4
Typical error in calculating the mean of an underlying process by using a finite number of samples, N. The parameter, s , is the standard deviation.

$$\textit{Typical error} = \frac{\sigma}{N^{1/2}}$$

If N is small, the statistical noise in the calculated mean will be very large. In other words, you do not have access to enough data to properly characterize the process. The larger the value of N, the smaller the expected error will become. A milestone in probability theory, the *Strong Law of Large Numbers*, guarantees that the error becomes zero as N approaches infinity.

In the next step, we would like to calculate the standard deviation of the acquired signal, and use it as an estimate of the standard deviation of the underlying process. Herein lies the problem. Before you can calculate the standard deviation using Eq. 2-2, you need to already know the mean, μ. However, you don't know the mean of the underlying process, only the mean of the N point signal, *which contains an error due to statistical noise*. This error tends to reduce the calculated value of the standard deviation. To compensate for this, N is replaced by N-1. If N is large, the difference doesn't matter. If N is small, this replacement provides a more accurate